DRAFT: Optimal Harvesting in a Stochastic Predator–Prey System: Analytical Characterization and Numerical Resolution

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Abstract

We study the problem of optimally harvesting two interacting species—one a predator of the other—modeled using a stochastic Lotka–Volterra system with harvesting. The objective is to maximize total expected profit from fishing both species over a finite time horizon, accounting for nonlinear extraction costs and environmental stochasticity. We derive the necessary conditions for optimality using the stochastic maximum principle, in cases where the diffusion is and is not dependent on the control. Where analytical solutions are intractable, we employ a forward-backward sweep method to compute the optimal harvesting strategy numerically. The paper concludes with simulations and comparisons with linearized and deterministic approximations.

Contents

1 Introduction

Sustainable management of renewable biological resources—especially fisheries—has long been a central challenge in both ecological and economic modeling. Classical approaches have addressed optimal harvesting for single species under deterministic assumptions, often using tools from calculus of variations or optimal control. However, such models neglect important features of real-world ecosystems, including interspecies interactions and environmental uncertainty.

In this paper, we consider a more realistic setting: a predator–prey fishery in which both species are subject to harvesting, and their populations evolve stochastically due to environmental variability. Specifically, we model the ecosystem with a stochastic version of the classical Lotka–Volterra system, incorporating both ecological dynamics and economic incentives. The goal is to determine optimal harvesting strategies for the predator and prey populations that maximize expected profit over a finite time horizon. We account for price differences between species and include nonlinear harvesting costs that increase when populations are small or harvest rates are high.

A novel feature of this study is the comparison between two models of environmental noise: one in which the diffusion term (representing stochastic fluctuations) is independent of the harvesting effort, and another in which the diffusion explicitly depends on the control. The latter captures scenarios where intense harvesting increases ecosystem variability, either through habitat disruption or destabilizing biological interactions.

To solve each problem, we employ the stochastic maximum principle (SMP) to derive necessary conditions for optimality and formulate the associated forward-backward stochastic differential equations (FBSDEs). We then implement the forward-backward sweep method to solve these systems numerically and simulate optimal population and control trajectories under both diffusion assumptions.

This study contributes a comparative analysis of two widely relevant modeling approaches in stochastic ecological control. By quantifying the effects of control-dependent diffusion, we provide insight into when more complex models are justified and how they influence optimal harvesting decisions.

2 Model Formulation

We consider a finite-horizon stochastic control problem in a two-species fishery where one species is a natural predator of the other. Let $x_1(t)$ and $x_2(t)$ denote the population densities of the prey and predator species, respectively, at time $t \in [0, T]$. The baseline dynamics are based on the well-known Lotka–Volterra (LV) model [2, 6], which captures predator-prey interactions through the system:

$$\frac{dx_1}{dt} = \alpha x_1 - \beta x_1 x_2 \tag{1}$$

$$\frac{dx_2}{dt} = -\gamma x_2 + \delta x_1 x_2 \tag{2}$$

where α and γ represent intrinsic growth and death rates of the prey and predator species, respectively, and β, δ represent interaction coefficients. All parameters are assumed positive.

This model assumes that prey reproduce exponentially in the absence of predation and that the interaction rate is proportional to the product of the species populations. However, real ecosystems are subject to:

- **Harvesting by humans**, which depletes population levels.
- **Environmental uncertainty**, which leads to stochastic fluctuations in population dynamics.
- **Economic incentives**, including species prices and harvesting costs.

We now introduce harvesting controls and stochasticity into the model, and consider two formulations:

2.1 Model A: Control-Independent Diffusion

In this formulation, population fluctuations are purely environmental and not directly influenced by harvesting intensity. The population dynamics are governed by:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t) x_2(t) - u_1(t)]\, dt + \sigma_1 x_1(t)\, dW_1(t) \tag{3}$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t) x_2(t) - u_2(t)]\, dt + \sigma_2 x_2(t)\, dW_2(t) \tag{4}$$

where:

- $u_1(t), u_2(t) \geq 0$ are the harvesting rates for prey and predator.
- $W_1(t)$ and $W_2(t)$ are independent standard Wiener processes.
- σ_1, σ_2 quantify the intensity of environmental variability.

2.2 Model B: Control-Dependent Diffusion

Here we allow harvesting intensity to influence environmental variability, modeling the idea that aggressive harvesting may destabilize ecosystems or introduce additional uncertainty. The dynamics become:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)]\,dt + (\sigma_{10}x_1(t) + \sigma_{11}u_1(t))\,dW_1(t) \tag{5}$$

$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)]\,dt + (\sigma_{20}x_2(t) + \sigma_{22}u_2(t))\,dW_2(t) \tag{6}$$

where the terms $\sigma_{11}u_1(t)$ and $\sigma_{22}u_2(t)$ represent additional volatility induced by harvesting activity.

2.3 Objective Functional

For both models, the controller aims to maximize the expected net profit:

$$J(u_1, u_2) = \mathbb{E}\left[\int_0^T \left(F_1u_1(t) + F_2u_2(t) - \frac{\gamma_1u_1(t)^2}{x_1(t)} - \frac{\gamma_2u_2(t)^2}{x_2(t)}\right) dt\right] \tag{7}$$

where

- $F_1, F_2 > 0$ are the market prices for prey and predator species.
- $\gamma_1, \gamma_2 > 0$ scale the nonlinear harvesting costs, which increase when populations are small or effort is high.

2.4 Admissibility and Constraints

The controls $u_1(t), u_2(t)$ must be non-negative, adapted to the natural filtration of $\{W_1(s), W_2(s)\}_{s\leq t}$, and satisfy integrability conditions ensuring that $J(u_1, u_2)$ is well-defined. State and control variables must satisfy:

$$x_1(t) > 0, \quad x_2(t) > 0, \quad u_1(t) \geq 0, \quad u_2(t) \geq 0, \quad \forall t \in [0, T]. \tag{8}$$

In the next section, we derive the necessary optimality conditions for both models using the stochastic maximum principle. This includes the construction of Hamiltonians, derivation of adjoint equations, and characterization of the optimal controls.

3 Stochastic Maximum Principle and Optimality Conditions

To characterize the optimal harvesting strategies $(u_1^*(t), u_2^*(t))$, we apply the stochastic maximum principle (SMP), which provides necessary conditions for optimality in controlled Itô diffusion systems. We follow the formulation in [7], focusing on the case where the drift depends on the control. In this section, we consider two settings: Model A, in which the diffusion term is independent of the control, and Model B, where it is control-dependent.

Let the net profit rate be denoted by $f(x_1, x_2, u_1, u_2)$ and suppose there is no terminal reward, denoted by $g(x_1(T), x_2(T)) = 0$. The objective is to maximize the expected total profit:

$$J(U) = \mathbb{E}\left[\int_0^T f(X(t), U(t))\,dt\right] \tag{9}$$

subject to the stochastic dynamics of the population densities. For generality, we write the system as:

$$dX(t) = b(X(t), U(t))\, dt + \Sigma(X(t), U(t))\, dW(t), \tag{10}$$

where:

- $X(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$ is the state vector of prey and predator populations.
- $U(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$ is the control vector of harvesting rates.
- $W(t) = \begin{bmatrix} W_1(t) \\ W_2(t) \end{bmatrix}$ is a 2-dimensional standard Brownian motion.
- $b(X, U) = \begin{bmatrix} \alpha x_1 - \beta x_1 x_2 - u_1 \\ -\gamma x_2 + \delta x_1 x_2 - u_2 \end{bmatrix}$ is the drift term, which is the same for both models
- $\Sigma : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^{2\times 2}$ is the diffusion matrix, which differs between models A and B.

Additionally, the profit function is:

$$f(X, U) = F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2}. \tag{11}$$

The stochastic maximum principle introduces an adjoint process $(P(t), Q(t))$ satisfying a backward stochastic differential equation (BSDE):

$$dP(t) = -\left[\nabla_x b(X, U)^\top P(t) + \sum_{j=1}^{2} \nabla_x \sigma^j (X, U)^\top Q_j(t) - \nabla_x f(X, U) \right] dt + Q(t) dW(t), \tag{12}$$

where σ^j denotes the j-th column of the diffusion matrix Σ and $Q(t)$ is a 2×2 matrix whose j-th column is $Q_j(t)$.

3.1 Model A: Control-Independent Diffusion

In this case, we take $\Sigma(X)$ to be a diagonal matrix:

$$\Sigma(X) = \begin{bmatrix} \sigma_1 x_1 & 0 \\ 0 & \sigma_2 x_2 \end{bmatrix}, \qquad \text{with } \sigma_1, \sigma_2 > 0.$$

Let $P(t) = \begin{bmatrix} p_1(t) \\ p_2(t) \end{bmatrix}$ and $Q(t) = \begin{bmatrix} q_1(t) & 0 \\ 0 & q_2(t) \end{bmatrix}$. Then the adjoint equations are:

$$dp_1(t) = -\left[(\alpha - \beta x_2) p_1 + \delta x_2 p_2 + \sigma_1 q_1 - \frac{\gamma_1 u_1^2}{x_1^2} \right] dt + q_1\, dW_1(t), \tag{13}$$

$$dp_2(t) = -\left[-\beta x_1 p_1 + (-\gamma + \delta x_1) p_2 + \sigma_2 q_2 - \frac{\gamma_2 u_2^2}{x_2^2} \right] dt + q_2\, dW_2(t), \tag{14}$$

with terminal conditions $p_1(T) = 0$, $p_2(T) = 0$. The Hamiltonian for this model is:

$$
\begin{aligned}
H(X, U, P, Q) &= f(X, U) + P^\top b(X, U) + \operatorname{tr}(Q^\top \Sigma(X)) && (15) \\
&= F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2} \\
&\quad + p_1(\alpha x_1 - \beta x_1 x_2 - u_1) + p_2(-\gamma x_2 + \delta x_1 x_2 - u_2) \\
&\quad + q_1 \sigma_1 x_1 + q_2 \sigma_2 x_2 && (16)
\end{aligned}
$$

Since the diffusion is independent of control, the SMP condition for optimality is:

$$
H(X^*, U^*, P, Q) = \max_{U \in \mathcal{U}} H(X, U, P, Q). \tag{17}
$$

Setting the derivatives $\partial H / \partial u_i = 0$, we find the optimal controls (assuming interior maxima):

$$
\frac{\partial H}{\partial u_1} = F_1 - \frac{2\gamma_1 u_1}{x_1} - p_1 = 0 \quad \Rightarrow \quad u_1^* = \frac{x_1}{2\gamma_1}(F_1 - p_1) \tag{18}
$$

$$
\frac{\partial H}{\partial u_2} = F_2 - \frac{2\gamma_2 u_2}{x_2} - p_2 = 0 \quad \Rightarrow \quad u_2^* = \frac{x_2}{2\gamma_2}(F_2 - p_2) \tag{19}
$$

These expressions define the optimal harvesting policy in terms of the current population and the adjoint process values. While the adjoint processes $p_1(t), p_2(t)$ are governed by backward stochastic differential equations (BSDEs), the population dynamics in equation (10) define forward stochastic processes. Substituting the optimal controls

$$
U^*(t) = \begin{bmatrix} u_1^*(t) \\ u_2^*(t) \end{bmatrix}
$$

into the system yields a coupled system of forward-backward stochastic differential equations (FBSDEs), which can be solved numerically to obtain the optimal harvesting strategy.

3.2 Model B: Control-Dependent Diffusion

In this case, we allow the diffusion coefficients to depend explicitly on the control variables, capturing additional variability due to harvesting intensity. The diffusion matrix $\Sigma(X, U)$ is still diagonal, but now has control-dependent entries:

$$
\Sigma(X, U) = \begin{bmatrix} \sigma_{10} x_1 + \sigma_{11} u_1 & 0 \\ 0 & \sigma_{20} x_2 + \sigma_{22} u_2 \end{bmatrix}. \tag{20}
$$

Recall the forward dynamics are now:

$$
dx_1(t) = [\alpha x_1(t) - \beta x_1(t) x_2(t) - u_1(t)]\, dt + (\sigma_{10} x_1(t) + \sigma_{11} u_1(t))\, dW_1(t), \tag{21}
$$

$$
dx_2(t) = [-\gamma x_2(t) + \delta x_1(t) x_2(t) - u_2(t)]\, dt + (\sigma_{20} x_2(t) + \sigma_{22} u_2(t))\, dW_2(t). \tag{22}
$$

We use the same first-order adjoint process $(P(\cdot), Q(\cdot))$ described by (12). For this model, we have

$$
dp_1(t) = -\left[(\alpha - \beta x_2) p_1 + \delta x_2 p_2 + \sigma_{10} q_1 - \frac{\gamma_1 u_1^2}{x_1^2}\right] dt + q_1\, dW_1(t), \tag{23}
$$

$$
dp_2(t) = -\left[-\beta x_1 p_1 + (-\gamma + \delta x_1) p_2 + \sigma_{20} q_2 - \frac{\gamma_2 u_2^2}{x_2^2}\right] dt + q_2\, dW_2(t), \tag{24}
$$

To account for control-dependent noise, we now introduce a second-order adjoint process, solved by $(R(\cdot), S(\cdot))$. Let:

$$R(t) = \begin{bmatrix} r_{11}(t) & r_{12}(t) \\ r_{21}(t) & r_{22}(t) \end{bmatrix}, \qquad S_j(t) = \begin{bmatrix} s_{11}^{(j)}(t) & s_{12}^{(j)}(t) \\ s_{21}^{(j)}(t) & s_{22}^{(j)}(t) \end{bmatrix} \quad \text{for } j = 1, 2.$$

These processes satisfy a second-order backward stochastic differential equation,

$$dR(t) = -\left[b_x^\top R + R b_x + \sum_{j=1}^{2} \sigma_x^{j\top} R \sigma_x^j + \sum_{j=1}^{2} \left(\sigma_x^{j\top} S_j + S_j \sigma_x^j \right) + H_{xx} \right] dt + \sum_{j=1}^{2} S_j(t) dW_j(t), \tag{25}$$

where b_x and σ_x^j denote the Jacobians of the drift and diffusion coefficients with respect to the state variables X, and H is the Hamiltonian defined in (15). The BSDE has the terminal condition:

$$R(T) = -g_{xx}(X^*(T)) = 0.$$

Solving for the dynamics of the second-order adjoint process, we find:

$$\begin{aligned} dR(t) = -\Bigg\{ & \begin{bmatrix} \alpha - \beta x_2 & \delta x_2 \\ -\beta x_1 & -\gamma + \delta x_1 \end{bmatrix}^\top R(t) + R(t) \begin{bmatrix} \alpha - \beta x_2 & \delta x_2 \\ -\beta x_1 & -\gamma + \delta x_1 \end{bmatrix} \\ & + \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix}^\top R(t) \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix}^\top R(t) \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix} \\ & + \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix}^\top S_1(t) + S_1(t) \begin{bmatrix} \sigma_{10} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix}^\top S_2(t) + S_2(t) \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} \end{bmatrix} \\ & + \begin{bmatrix} 2\gamma_1 \dfrac{u_1^2}{x_1^3} & 0 \\ 0 & 2\gamma_2 \dfrac{u_2^2}{x_2^3} \end{bmatrix} \Bigg\} dt + S_1(t)\, dW_1(t) + S_2(t)\, dW_2(t), \end{aligned} \tag{26}$$

To account for the added uncertainty introduced by control-dependent diffusion, we adopt a generalized Hamiltonian function G as formulated in [7], which includes a second-order risk adjustment term:

$$G(X,U,P,R) = f(X,U) + P^\top b(X,U) + \frac{1}{2} \operatorname{tr} \left[\Sigma(X,U)^\top R \Sigma(X,U) \right], \; (X,U,P,R) \in \mathbb{R}^2 \times \mathcal{U} \times \mathbb{R}^2 \times \mathcal{S}^2, \tag{27}$$

where $\mathcal{U}$ is the set of admissible controls and $\mathcal{S}^2$ is the set of 2×2 symmetric matrices.

Unlike the SMP used in Model A, the stochastic maximum principle for systems with control-dependent diffusion does not maximize G directly. Instead, it guarantees pairs of adjoint processes that maximize an auxiliary function $\mathcal{H}$, defined as:

$$\mathcal{H}(X,U) = G(X,U,P,R) + \operatorname{tr} \left[\Sigma(X,U)^\top (Q - R\Sigma(X,U)) \right], \tag{28}$$

which is associated with an optimal 6-tuple (X^*, U^*, P, Q, R, S).

Expanding, we obtain

$$\begin{aligned} \mathcal{H}(X,U) &= f(X,U) + P^\top b(X,U) + \frac{1}{2} \operatorname{tr} \left[\Sigma^\top R \Sigma \right] + \operatorname{tr} \left[\Sigma^\top Q - \Sigma^\top R \Sigma \right] & (29) \\ &= f(X,U) + P^\top b(X,U) + \operatorname{tr} \left[\Sigma^\top Q \right] - \frac{1}{2} \operatorname{tr} \left[\Sigma^\top R \Sigma \right]. & (30) \end{aligned}$$

We now compute $\mathcal{H}(X, U)$ explicitly. Recall:

$$f(X, U) = F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2},$$

$$b(X, U) = \begin{bmatrix} \alpha x_1 - \beta x_1 x_2 - u_1 \\ -\gamma x_2 + \delta x_1 x_2 - u_2 \end{bmatrix}, \quad P = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}, \quad Q = \begin{bmatrix} q_1 & 0 \\ 0 & q_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_{10} x_1 + \sigma_{11} u_1 & 0 \\ 0 & \sigma_{20} x_2 + \sigma_{22} u_2 \end{bmatrix}.$$

Substituting these into $\mathcal{H}$, we obtain:

$$\begin{aligned} \mathcal{H}(X, U) = {} & F_1 u_1 + F_2 u_2 - \frac{\gamma_1 u_1^2}{x_1} - \frac{\gamma_2 u_2^2}{x_2} \\ & + p_1(\alpha x_1 - \beta x_1 x_2 - u_1) + p_2(-\gamma x_2 + \delta x_1 x_2 - u_2) \\ & + q_1(\sigma_{10} x_1 + \sigma_{11} u_1) + q_2(\sigma_{20} x_2 + \sigma_{22} u_2) \\ & - \frac{1}{2}\left[r_{11}(\sigma_{10} x_1 + \sigma_{11} u_1)^2 + r_{22}(\sigma_{20} x_2 + \sigma_{22} u_2)^2 \right]. \end{aligned} \tag{31}$$

To find the optimal control, we compute the first-order conditions by differentiating the Hamiltonian $\mathcal{H}(X, U)$ with respect to each control u_i and setting the result to zero:

$$\frac{\partial \mathcal{H}}{\partial u_1} = F_1 - \frac{2\gamma_1 u_1}{x_1} - p_1 + q_1 \sigma_{11} - r_{11}(\sigma_{10} x_1 + \sigma_{11} u_1)\sigma_{11}, \tag{32}$$

$$\frac{\partial \mathcal{H}}{\partial u_2} = F_2 - \frac{2\gamma_2 u_2}{x_2} - p_2 + q_2 \sigma_{22} - r_{22}(\sigma_{20} x_2 + \sigma_{22} u_2)\sigma_{22}. \tag{33}$$

Setting these derivatives equal to zero yields the necessary optimality conditions:

$$0 = F_1 - \frac{2\gamma_1 u_1}{x_1} - p_1 + q_1 \sigma_{11} - r_{11}(\sigma_{10} x_1 + \sigma_{11} u_1)\sigma_{11}, \tag{34}$$

$$0 = F_2 - \frac{2\gamma_2 u_2}{x_2} - p_2 + q_2 \sigma_{22} - r_{22}(\sigma_{20} x_2 + \sigma_{22} u_2)\sigma_{22}. \tag{35}$$

As expected, the optimal controls for this model are much more complex than the control-independent diffusion model. Analytic solutions are almost sure to be impossible to find, so we proceed with a numerical approach.

4 Numerical Solution via Forward-Backward Sweep

There are a few approaches to solving FBSDEs numerically [1, 3]. Here, a numerical solution of the stochastic optimal control problem for the predator-prey fishery models is achieved using the *forward-backward sweep method*, an iterative technique designed to solve coupled FBSDEs. This method is well-suited for optimal control problems governed by SDEs, where the state dynamics evolve forward in time and the adjoint processes, derived earlier, evolve backward. The forward-backward sweep method alternates between computing the state trajectories forward in time, solving the adjoint equations backward, and updating the control variables to satisfy the optimality conditions. By iterating these steps, the method converges to the optimal control strategy $(u_1^*(t), u_2^*(t))$, accounting for the stochastic nature of the system through Monte Carlo simulations and ensuring adaptedness to the filtration $\mathcal{F}_t = \sigma(W_1(s), W_2(s) : s \leq t)$. Because of the high degrees of nonlinearity and sensitivity of the models, a weighted update of the controls during the sweep is performed, as in [4, 5], adding a convex combination of the new control values and previous values.

4.1 Model Equations

The system parameters are: intrinsic growth rate of prey $\alpha = 0.5$, predation rate $\beta = 0.02$, natural mortality rate of predator $\gamma = 0.4$, conversion efficiency $\delta = 0.01$, diffusion coefficients $\sigma_1 = \sigma_2 = 0.1$ (Model A), $\sigma_{10} = \sigma_{20} = 0.1$, $\sigma_{11} = \sigma_{22} = 0.05$ (Model B), harvesting profits $F_1 = 10$, $F_2 = 20$, and cost coefficients $\gamma_1 = 50$, $\gamma_2 = 100$. The initial conditions are $x_1(0) = 100$, $x_2(0) = 20$, and the time horizon is $T = 100$ days, roughly representing a fishing season.

Recall the forward SDEs governing the state dynamics:

- **Model A (Control-Independent Diffusion)**:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)]\, dt + \sigma_1 x_1(t)\, dW_1(t), \tag{36}$$
$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)]\, dt + \sigma_2 x_2(t)\, dW_2(t). \tag{37}$$

- **Model B (Control-Dependent Diffusion)**:

$$dx_1(t) = [\alpha x_1(t) - \beta x_1(t)x_2(t) - u_1(t)]\, dt + (\sigma_{10} x_1(t) + \sigma_{11} u_1(t))\, dW_1(t), \tag{38}$$
$$dx_2(t) = [-\gamma x_2(t) + \delta x_1(t)x_2(t) - u_2(t)]\, dt + (\sigma_{20} x_2(t) + \sigma_{22} u_2(t))\, dW_2(t). \tag{39}$$

The first-order adjoint BSDEs, derived from the SMP, are:

- **Model A**:

$$dp_1(t) = -\left[(\alpha - \beta x_2(t))p_1(t) + \delta x_2(t)p_2(t) + \sigma_1 q_1(t) - \frac{\gamma_1 u_1(t)^2}{x_1(t)^2}\right] dt + q_1(t)\, dW_1(t), \tag{40}$$
$$dp_2(t) = -\left[-\beta x_1(t)p_1(t) + (-\gamma + \delta x_1(t))p_2(t) + \sigma_2 q_2(t) - \frac{\gamma_2 u_2(t)^2}{x_2(t)^2}\right] dt + q_2(t)\, dW_2(t), \tag{41}$$

  with terminal conditions $p_1(T) = p_2(T) = 0$.

- **Model B**:

$$dp_1(t) = -\left[(\alpha - \beta x_2(t))p_1(t) + \delta x_2(t)p_2(t) + \sigma_{10} q_1(t) - \frac{\gamma_1 u_1(t)^2}{x_1(t)^2}\right] dt + q_1(t)\, dW_1(t), \tag{42}$$
$$dp_2(t) = -\left[-\beta x_1(t)p_1(t) + (-\gamma + \delta x_1(t))p_2(t) + \sigma_{20} q_2(t) - \frac{\gamma_2 u_2(t)^2}{x_2(t)^2}\right] dt + q_2(t)\, dW_2(t), \tag{43}$$

  with terminal conditions $p_1(T) = p_2(T) = 0$.

For Model B, the second-order adjoint BSDE for the matrix-valued process $R(t) = \begin{bmatrix} r_{11}(t) & r_{12}(t) \\ r_{21}(t) & r_{22}(t) \end{bmatrix}$ is:

$$dR(t) = -\left[b_x^\top R(t) + R(t)b_x + \sum_{j=1}^{2} \sigma_x^{j\top} R(t)\sigma_x^j + \sum_{j=1}^{2} \left(\sigma_x^{j\top} S_j(t) + S_j(t)\sigma_x^j\right) + H_{xx}\right] dt + \sum_{j=1}^{2} S_j(t) dW_j(t), \tag{44}$$

where:

$$b_x = \begin{bmatrix} \alpha - \beta x_2 & -\beta x_1 \\ \delta x_2 & -\gamma + \delta x_1 \end{bmatrix}, \quad \sigma_x^1 = \begin{bmatrix} \sigma_{10} + \sigma_{11}\frac{u_1}{x_1} & 0 \\ 0 & 0 \end{bmatrix}, \quad \sigma_x^2 = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_{20} + \sigma_{22}\frac{u_2}{x_2} \end{bmatrix}, \quad H_{xx} = \begin{bmatrix} 2\gamma_1 \frac{u_1^2}{x_1^3} & 0 \\ 0 & 2\gamma_2 \frac{u_2^2}{x_2^3} \end{bmatrix},$$

and $S_j(t) = \begin{bmatrix} s_{11}^{(j)}(t) & s_{12}^{(j)}(t) \\ s_{21}^{(j)}(t) & s_{22}^{(j)}(t) \end{bmatrix}$, with terminal condition $R(T) = 0$. The diagonal components are:

$$dr_{11}(t) = -\left[2(\alpha - \beta x_2)r_{11} - \beta x_1 r_{12} + \delta x_2 r_{21} + \sigma_{10}^2 r_{11} + 2\sigma_{10} s_{11}^{(1)} + 2\gamma_1 \frac{u_1^2}{x_1^3}\right] dt + s_{11}^{(1)} dW_1(t) + s_{11}^{(2)} dW_2(t), \tag{45}$$

$$dr_{22}(t) = -\left[-\beta x_1 r_{12} + 2(-\gamma + \delta x_1)r_{22} + \delta x_2 r_{21} + \sigma_{20}^2 r_{22} + 2\sigma_{20} s_{22}^{(2)} + 2\gamma_2 \frac{u_2^2}{x_2^3}\right] dt + s_{22}^{(1)} dW_1(t) + s_{22}^{(2)} dW_2(t). \tag{46}$$

The optimal controls are derived from the SMP. For Model A:

$$u_1^*(t) = \max\left(0, \frac{x_1(t)(F_1 - p_1(t))}{2\gamma_1}\right), \tag{47}$$

$$u_2^*(t) = \max\left(0, \frac{x_2(t)(F_2 - p_2(t))}{2\gamma_2}\right). \tag{48}$$

For Model B, the controls satisfy quadratic equations involving $p_j(t)$, $r_{jj}(t)$, and $s_{jj}^{(j)}(t)$, which are solved numerically at each time step to ensure optimality.

4.2 Numerical Algorithms

The time interval $[0, T]$ is discretized into $N = 10000$ points $t_i = i\Delta t$, where $\Delta t = T/N$. Monte Carlo simulations are performed over $M = 1000$ to 5000 sample paths, with Wiener increments approximated as $\Delta W_j(t_i) \approx \sqrt{\Delta t}\xi_{j,i}^{(m)}$, where $\xi_{j,i}^{(m)} \sim \mathcal{N}(0, 1)$. The stochastic components $q_j(t_i, m)$ and $s_{jj}^{(k)}(t_i, m)$ are estimated using conditional expectation via regression, employing polynomial basis functions $\{1, t_i, x_1(t_i, m), x_2(t_i, m), x_1(t_i, m)x_2(t_i, m), x_1(t_i, m)^2, x_2(t_i, m)^2\}$ to model their dependence on the state variables.

4.3 Numerical Considerations

The numerical scheme employs the Euler-Maruyama method for the forward SDEs and a backward Euler scheme for the BSDEs, ensuring stability for small Δt. The regression method approximates $q_j(t_i, m)$ and $s_{jj}^{(k)}(t_i, m)$ as linear combinations of basis functions, ensuring adaptedness to $\mathcal{F}_{t_i}$. A numerical threshold $\epsilon = 10^{-6}$ is used in denominators (e.g., $\frac{u_j^2}{x_j^2}$, $\frac{u_j^2}{x_j^3}$) to prevent division-by-zero errors. The number of sample paths M is chosen between 1000 and 5000 to balance accuracy and computational cost, with larger M reducing variance in the regression estimates.

Convergence of the forward-backward sweep is monitored by ensuring $\max_{i,m} |u_j^{\text{new}}(t_i, m) - u_j(t_i, m)| < 0.01$. The accuracy of the BSDE solutions is validated by checking the martingale property, i.e., $\sum_{m=1}^{M} p_j(t_{i-1}, m)\xi_{j,i}^{(m)} \approx 0$ and similarly for $r_{jj}(t_{i-1}, m)$. Regression residuals are also monitored to ensure the quality of the estimates for $q_j(t_i, m)$ and $s_{jj}^{(k)}(t_i, m)$.

Algorithm 1 Euler-Maruyama for $x_1(t)$, $x_2(t)$ (Model A)

Require: Initial conditions $x_1(0,m) = 100$, $x_2(0,m) = 20$, controls $u_1(t_i,m)$, $u_2(t_i,m)$, parameters, Δt, N, M

Ensure: State trajectories $x_1(t_i,m)$, $x_2(t_i,m)$

for each sample path $m = 1$ to M **do**

  Set $x_1(0,m) = 100$, $x_2(0,m) = 20$

  for $i = 0$ to $N-1$ **do**

    Generate $\xi_{1,i}^{(m)}, \xi_{2,i}^{(m)} \sim \mathcal{N}(0,1)$

    Compute drift:

$$f_1 = \alpha x_1(t_i,m) - \beta x_1(t_i,m)x_2(t_i,m) - u_1(t_i,m)$$

$$f_2 = -\gamma x_2(t_i,m) + \delta x_1(t_i,m)x_2(t_i,m) - u_2(t_i,m)$$

    Compute diffusion:

$$g_1 = \sigma_1 x_1(t_i,m), \quad g_2 = \sigma_2 x_2(t_i,m)$$

    Update:

$$x_1(t_{i+1},m) = x_1(t_i,m) + f_1\Delta t + g_1\sqrt{\Delta t}\xi_{1,i}^{(m)}$$

$$x_2(t_{i+1},m) = x_2(t_i,m) + f_2\Delta t + g_2\sqrt{\Delta t}\xi_{2,i}^{(m)}$$

    Ensure non-negativity: $x_1(t_{i+1},m) = \max(x_1(t_{i+1},m),\epsilon)$, $x_2(t_{i+1},m) = \max(x_2(t_{i+1},m),\epsilon)$, with $\epsilon = 10^{-6}$

  end for

end for

return $x_1(t_i,m)$, $x_2(t_i,m)$

Algorithm 2 Euler-Maruyama for $x_1(t)$, $x_2(t)$ (Model B)

Require: Initial conditions $x_1(0, m) = 100$, $x_2(0, m) = 20$, controls $u_1(t_i, m)$, $u_2(t_i, m)$, parameters, Δt, N, M

Ensure: State trajectories $x_1(t_i, m)$, $x_2(t_i, m)$

for each sample path $m = 1$ to M **do**

  Set $x_1(0, m) = 100$, $x_2(0, m) = 20$

  for $i = 0$ to $N - 1$ **do**

    Generate $\xi_{1,i}^{(m)}, \xi_{2,i}^{(m)} \sim \mathcal{N}(0, 1)$

    Compute drift:

$$f_1 = \alpha x_1(t_i, m) - \beta x_1(t_i, m) x_2(t_i, m) - u_1(t_i, m)$$

$$f_2 = -\gamma x_2(t_i, m) + \delta x_1(t_i, m) x_2(t_i, m) - u_2(t_i, m)$$

    Compute diffusion:

$$g_1 = \sigma_{10} x_1(t_i, m) + \sigma_{11} u_1(t_i, m), \quad g_2 = \sigma_{20} x_2(t_i, m) + \sigma_{22} u_2(t_i, m)$$

    Update:

$$x_1(t_{i+1}, m) = x_1(t_i, m) + f_1 \Delta t + g_1 \sqrt{\Delta t} \xi_{1,i}^{(m)}$$

$$x_2(t_{i+1}, m) = x_2(t_i, m) + f_2 \Delta t + g_2 \sqrt{\Delta t} \xi_{2,i}^{(m)}$$

    Ensure non-negativity: $x_1(t_{i+1}, m) = \max(x_1(t_{i+1}, m), \epsilon)$, $x_2(t_{i+1}, m) = \max(x_2(t_{i+1}, m), \epsilon)$

  end for

end for

return $x_1(t_i, m)$, $x_2(t_i, m)$

Algorithm 3 Backward Euler with Regression for $p_1(t)$, $p_2(t)$ (Model A)

Require: State trajectories $x_1(t_i, m)$, $x_2(t_i, m)$, controls $u_1(t_i, m)$, $u_2(t_i, m)$, Wiener increments $\xi_{1,i}^{(m)}$, $\xi_{2,i}^{(m)}$, parameters, Δt, N, M

Ensure: Adjoint trajectories $p_1(t_i, m)$, $p_2(t_i, m)$, estimates $q_1(t_i, m)$, $q_2(t_i, m)$

1: Initialize $p_1(t_N, m) = 0$, $p_2(t_N, m) = 0$, $q_1(t_i, m) = 0$, $q_2(t_i, m) = 0$ for all i, m

2: **for** $i = N - 1$ down to 0 **do**

3: **for** each sample path $m = 1$ to M **do**

4: Compute drift:

$$f_1 = (\alpha - \beta x_2(t_i, m))p_1(t_i, m) + \delta x_2(t_i, m)p_2(t_i, m) + \sigma_1 q_1(t_i, m) - \frac{\gamma_1 u_1(t_i, m)^2}{\max(x_1(t_i, m)^2, \epsilon)}$$

$$f_2 = -\beta x_1(t_i, m)p_1(t_i, m) + (-\gamma + \delta x_1(t_i, m))p_2(t_i, m) + \sigma_2 q_2(t_i, m) - \frac{\gamma_2 u_2(t_i, m)^2}{\max(x_2(t_i, m)^2, \epsilon)}$$

5: Update:

$$p_1(t_{i-1}, m) = p_1(t_i, m) + f_1 \Delta t + q_1(t_i, m)\sqrt{\Delta t}\xi_{1,i}^{(m)}$$

$$p_2(t_{i-1}, m) = p_2(t_i, m) + f_2 \Delta t + q_2(t_i, m)\sqrt{\Delta t}\xi_{2,i}^{(m)}$$

6: **end for**

7: Compute regression targets: $y_j = p_j(t_{i-1}, m)\xi_{j,i}^{(m)}$ for $j = 1, 2$

8: Form basis functions: $\phi_l(t_i, x_1(t_i, m), x_2(t_i, m)) = \{1, t_i, x_1, x_2, x_1 x_2, x_1^2, x_2^2\}$

9: Solve least-squares regression:

$$q_j(t_i, m) = \frac{\sum_l \beta_{j,l}\phi_l(t_i, x_1(t_i, m), x_2(t_i, m))}{\sqrt{\Delta t}}, \quad \min_{\beta_{j,l}} \sum_{m=1}^{M} \left| y_j - \sum_l \beta_{j,l}\phi_l(t_i, x_1(t_i, m), x_2(t_i, m)) \right|^2$$

10: **end for**

11: **return** $p_1(t_i, m)$, $p_2(t_i, m)$, $q_1(t_i, m)$, $q_2(t_i, m)$

Algorithm 4 Backward Euler with Regression for $r_{11}(t)$, $r_{22}(t)$ (Model B)

Require: State trajectories $x_1(t_i, m)$, $x_2(t_i, m)$, controls $u_1(t_i, m)$, $u_2(t_i, m)$, adjoints $r_{12}(t_i, m)$, $r_{21}(t_i, m)$, Wiener increments $\xi_{j,i}^{(m)}$, parameters, Δt, N, M

Ensure: Adjoint trajectories $r_{11}(t_i, m)$, $r_{22}(t_i, m)$, estimates $s_{11}^{(j)}(t_i, m)$, $s_{22}^{(j)}(t_i, m)$

Initialize $r_{11}(t_N, m) = 0$, $r_{22}(t_N, m) = 0$, $s_{11}^{(j)}(t_i, m) = 0$, $s_{22}^{(j)}(t_i, m) = 0$ for all i, m, $j = 1, 2$

for $i = N - 1$ down to 0 **do**

  for each sample path $m = 1$ to M **do**

    Compute drift for r_{11}:

$$
\begin{aligned}
f_{11} = {} & 2(\alpha - \beta x_2(t_i, m)) r_{11}(t_i, m) - \beta x_1(t_i, m) r_{12}(t_i, m) \\
& + \delta x_2(t_i, m) r_{21}(t_i, m) + \sigma_{10}^2 r_{11}(t_i, m) \\
& + 2\sigma_{10} s_{11}^{(1)}(t_i, m) + 2\gamma_1 \frac{u_1(t_i, m)^2}{\max(x_1(t_i, m)^3, \epsilon)}
\end{aligned}
$$

    Compute drift for r_{22}:

$$
\begin{aligned}
f_{22} = {} & -\beta x_1(t_i, m) r_{12}(t_i, m) + 2(-\gamma + \delta x_1(t_i, m)) r_{22}(t_i, m) \\
& + \delta x_2(t_i, m) r_{21}(t_i, m) + \sigma_{20}^2 r_{22}(t_i, m) \\
& + 2\sigma_{20} s_{22}^{(2)}(t_i, m) + 2\gamma_2 \frac{u_2(t_i, m)^2}{\max(x_2(t_i, m)^3, \epsilon)}
\end{aligned}
$$

    Update:

$$
r_{11}(t_{i-1}, m) = r_{11}(t_i, m) + f_{11}\Delta t + s_{11}^{(1)}(t_i, m)\sqrt{\Delta t}\xi_{1,i}^{(m)} + s_{11}^{(2)}(t_i, m)\sqrt{\Delta t}\xi_{2,i}^{(m)}
$$

$$
r_{22}(t_{i-1}, m) = r_{22}(t_i, m) + f_{22}\Delta t + s_{22}^{(1)}(t_i, m)\sqrt{\Delta t}\xi_{1,i}^{(m)} + s_{22}^{(2)}(t_i, m)\sqrt{\Delta t}\xi_{2,i}^{(m)}
$$

  end for

  Compute regression targets: $z_{jj}^{(k)} = r_{jj}(t_{i-1}, m)\xi_{k,i}^{(m)}$ for $j, k = 1, 2$

  Form basis functions: $\phi_l(t_i, x_1(t_i, m), x_2(t_i, m))$

  Solve least-squares regression:

$$
s_{jj}^{(k)}(t_i, m) = \frac{\sum_l \beta_{jj,l}^{(k)} \phi_l(t_i, x_1(t_i, m), x_2(t_i, m))}{\sqrt{\Delta t}}, \quad \min_{\beta_{jj,l}^{(k)}} \sum_{m=1}^{M} \left| z_{jj}^{(k)} - \sum_l \beta_{jj,l}^{(k)} \phi_l(t_i, x_1(t_i, m), x_2(t_i, m)) \right|^2
$$

end for

return $r_{11}(t_i, m)$, $r_{22}(t_i, m)$, $s_{11}^{(j)}(t_i, m)$, $s_{22}^{(j)}(t_i, m)$

Algorithm 5 Forward-Backward Sweep Method

Require: Initial controls $u_1(t_i, m) = 0$, $u_2(t_i, m) = 0$, parameters, Δt, N, M, tolerance $\epsilon_{\text{tol}} = 10^{-4}$, max iterations $K = 100$

Ensure: Optimal controls $u_1^*(t_i, m)$, $u_2^*(t_i, m)$, state and adjoint trajectories

for $k = 1$ to K **do**
  Compute $x_1(t_i, m)$, $x_2(t_i, m)$ using Algorithm 1 (Model A) or 2 (Model B)
  Compute $p_1(t_i, m)$, $p_2(t_i, m)$, $q_1(t_i, m)$, $q_2(t_i, m)$ using Algorithm 3
  if Model B **then**
    Compute $r_{11}(t_i, m)$, $r_{22}(t_i, m)$, $s_{11}^{(j)}(t_i, m)$, $s_{22}^{(j)}(t_i, m)$ using Algorithm 4
  end if
  Update controls:
  if Model A **then**

$$u_1^{\text{new}}(t_i, m) = \max\left(0, \frac{x_1(t_i, m)(F_1 - p_1(t_i, m))}{2\gamma_1}\right)$$

$$u_2^{\text{new}}(t_i, m) = \max\left(0, \frac{x_2(t_i, m)(F_2 - p_2(t_i, m))}{2\gamma_2}\right)$$

  else
    Solve quadratic equations for $u_1^{\text{new}}(t_i, m)$, $u_2^{\text{new}}(t_i, m)$ numerically
  end if
  Perform weighted update on $u_j^{\text{new}}(t_i, m)$
  if $\max_{i,m} |u_j^{\text{new}}(t_i, m) - u_j(t_i, m)| < \epsilon_{\text{tol}}$ for $j = 1, 2$ **then**
    Break
  end if
  Set $u_j(t_i, m) \leftarrow u_j^{\text{new}}(t_i, m)$
end for
return $u_1^*(t_i, m)$, $u_2^*(t_i, m)$, $x_j(t_i, m)$, $p_j(t_i, m)$, (Model B) $r_{jj}(t_i, m)$

5 Simulation Results

5.1 Model A

6 Discussion

7 Conclusion and Future Work

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